

Sensitization and habituation regulate reinforcer effectiveness

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ABSTRACT

We argue that sensitization and habituation occur to the sensory properties of reinforcers when those reinforcers are presented repeatedly or for a prolonged time. Sensitization increases, and habituation decreases, the ability of a reinforcer to control behavior. Supporting this argument, the rate of operant responding changes systematically within experimental sessions even when the programmed rate of reinforcement is held constant across the session. These within-session changes in operant responding are produced by repeated delivery of the reinforcer, and their empirical characteristics correspond to the characteristics of behavior undergoing sensitization and habituation. Two characteristics of habituation (dishabituation, stimulus specificity) are particularly useful in separating habituation from alternative explanations. Arguing that habituation occurs to reinforcers expands the domain of habituation. The argument implies that habituation occurs to biologically important, not just to neutral, stimuli. The argument also implies that habituation may be observed in “voluntary” (operant), not just in reflexive, behavior. Expanding the domain of habituation has important implications for understanding operant and classical conditioning. Habituation may also contribute to the regulation of motivated behaviors. Habituation provides a more accurate and a less cumbersome explanation for motivated behaviors than homeostasis. Habituation also has some surprising, and easily testable, implications for the control of motivated behaviors.

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1. Introduction

We argue that sensitization and habituation occur to the sensory properties of reinforcers when those reinforcers are presented repeatedly or for a prolonged time. Reinforcers are stimuli that increase the frequency of responses that they follow (i.e., behavior is altered by its consequences). Habituation is a decrease in responsiveness to a stimulus that is presented repeatedly or for a prolonged time (e.g., Thompson & Spencer, 1966). Sensitization, the companion process of habituation, is an increase in responsiveness to a stimulus (e.g., Groves & Thompson, 1970). Although habituation, not sensitization, is the focus of the present discussion, it may be worth noting that sensitization usually occurs during the first few presentations of a stimulus or after the presentation of another strong stimulus. We argue that sensitization increases, and habituation decreases, the effectiveness of reinforcers in controlling behavior when those reinforcers are presented repeatedly or for a prolonged time.

Our argument expands the domain of sensitization and habituation in two ways. First, most studies of habituation examine the waning of a reflexive response (e.g., a leg flexion, Thompson & Spencer, 1966). Arguing that habituation occurs to reinforcers,

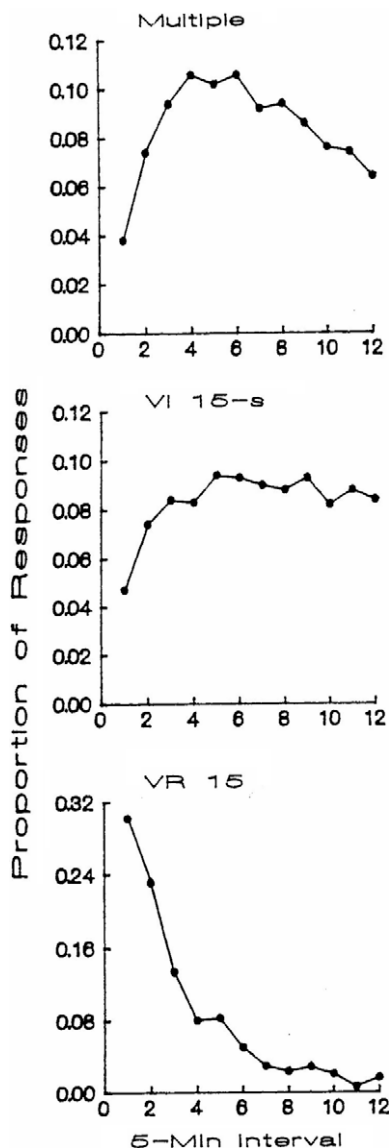
means that habituation may also be observed in “voluntary” or “goal-directed” operant behavior, not just in stimulus-elicited, reflexive, behavior. Second, most studies of habituation present a neutral stimulus, such as a light or tone (e.g., Harris, 1943; Thompson & Spencer, 1966). Some authors even assume that habituation does not occur to the repeated presentation of biologically significant stimuli (e.g., Thorpe, 1966). Arguing that sensitization and habituation occurs to reinforcers means that habituation also occurs to many biologically significant stimuli that are needed for the survival of the animal or the species (e.g., food, water).

2. Operant responding changes within experimental sessions

The idea that animals sensitize and then habituate to the sensory properties of reinforcers started from the observation that the rate of operant responding supported by a reinforcer (e.g., food, water) often changes systematically within experimental sessions. Rate of responding changes even when the conditions of reinforcement are held constant across the session (e.g., McSweeney, 1992; McSweeney, Weatherly, & Swindell, 1996a). Rate of operant responding often increases to a peak and then decreases, but it sometimes only increases or only decreases within the session (see Fig. 1). These within-session changes occur in steady state behavior. They are not transitional effects (e.g., acquisition curves). Within-session changes in operant responding are large, easily rep-

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Fig. 1. Three common within-session patterns of operant responding in nonhuman subjects responding for food reinforcers. Results are presented as proportions of total-session operant responses during successive 5-min intervals in 60-min experimental sessions for the mean of all subjects. Each graph presents the results when subjects responded on a different schedule of reinforcement. The results show that operant response rate may increase (middle graph), decrease (bottom graph), or increase and then decrease (top graph) within the experimental session even when the conditions of reinforcement are held constant across the session. From McSweeney et al. (1996, p. 257).

licable, highly general and not anticipated by existing theories of operant responding (e.g., McSweeney & Roll, 1993).

3. Competing explanations for within-session changes in responding

Several broadly stated potential explanations for within-session changes in operant responding were tested and rejected. Rejected explanations include: recovery from handling (McSweeney & Johnson, 1994), anticipation of events that follow the session (e.g., feeding or handling, McSweeney, Weatherly, & Swindell, 1995a), changes in a general motivational state (e.g., arousal, McSweeney, Swindell, & Weatherly, 1996a, 1996b), changes in interference from adjunctive behaviors (McSweeney et al., 1996a) or explora-

tion (Roll & McSweeney, 1997), changes in factors produced by the act of responding (e.g., muscular warmup, fatigue; McSweeney, Weatherly, & Roll, 1995; McSweeney, Weatherly, Roll, & Swindell, 1995; Melville, Rybiski, & Kamrani, 1996; Weatherly, McSweeney, & Swindell, 1995), and changes in “attention” to the operant task defined in several ways (McSweeney, Roll, & Weatherly, 1994; McSweeney, Weatherly, & Swindell, 1996b; Melville & Weatherly, 1996). An opponent-process explanation (e.g., Solomon & Corbit, 1974) also seems unlikely because early-session increases in responding sometimes occur without late-session decreases and vice versa.

Instead, within-session changes in operant responding are produced primarily by changes in the ability of the reinforcer to support behavior as a result of its repeated delivery. Altering characteristics of the reinforcer alters the within-session response pattern. For example, faster rates of reinforcer delivery yield steeper within-session decreases in responding than do slower rates (e.g., McSweeney, 1992). Probe preference tests also show that the ability of the reinforcer to support behavior changes systematically with repeated reinforcer delivery (McSweeney et al., 1996a).

4. Sensitization and habituation alter reinforcer effectiveness

Because food served as the reinforcer in most of our experiments, we initially hypothesized that satiation causes the within-session changes in reinforcer effectiveness. In fact, the definition of satiation that is usually used in the operant literature identifies these changes as satiation. For example, Millenson (1967, p. 367) states, “Satiation is...repeatedly presenting the reinforcer until it loses its power to reinforce.” Unfortunately, this definition of satiation provides a redundant description, not an explanation, for the within-session changes in operant responding. To avoid this problem, we turned to the satiety literature for a definition of satiation.

Those who study satiation often use that term to refer to the factors that contribute to the termination of ingestive behaviors such as feeding and drinking (e.g., Mook, 1996). Past research focused largely on post-ingestive factors (e.g., stomach distension, blood sugar levels, cholecystokinin [CCK] release). When a list of empirically determined satiety factors was used to predict the characteristics of within-session changes in responding, these predictions were incorrect. For example, manipulating such classic satiety factors as caloric density (Roll, McSweeney, Johnson, & Weatherly, 1995, Experiment 1), stomach loading (Roll et al., 1995, Experiment 3), reinforcer size (Temple, Giacomelli, Roem-mich, & Epstein, 2008; see Fig. 2), and deprivation (Roll et al., 1995, Experiment 2) failed to alter within-session response patterns. Within-session decreases in operant responding also occur even when subjects do not consume the reinforcer (Temple et al., 2008; see Fig. 2. This figure will be discussed further under Section 6.).

One satiety factor remained untested, however. Habituation to the sensory properties of food contributes to satiety for food (e.g., Epstein, Rodefer, Wisniewski, & Caggiula, 1992; Swithers & Hall, 1994). This led to the hypothesis that sensitization and habituation to the sensory properties of the reinforcer alter the effectiveness of that reinforcer in controlling behavior. According to this idea, both sensitization and habituation occur to reinforcers throughout the session as those reinforcers are repeatedly presented. Sensitization dominates during the first few reinforcer presentations. It increases the effectiveness of the reinforcer and is primarily responsible for the early-session increases in operant responding. Habituation dominates later, after several reinforcers have been presented. It decreases the effectiveness of the reinforcer and is primarily responsible for the late-session decreases in responding.

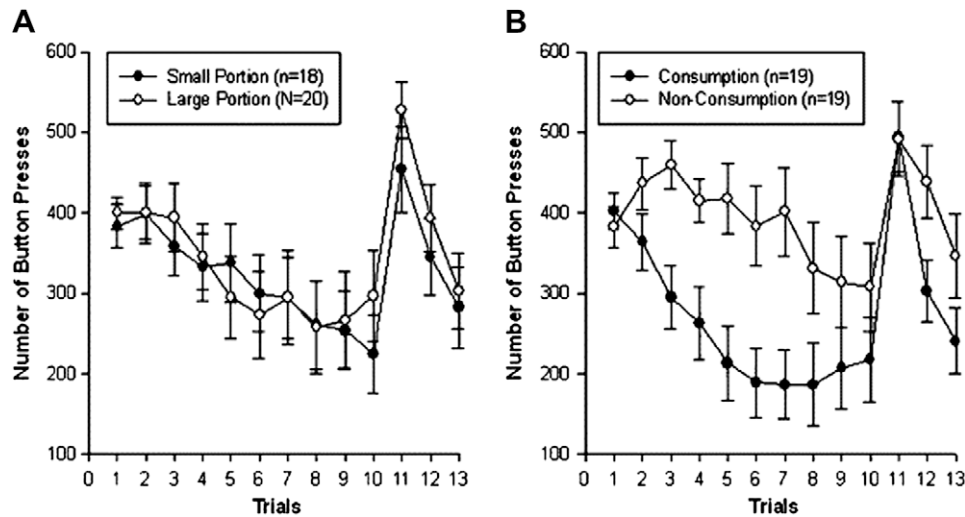


Fig. 2. Stimulus specificity of operant responding in humans. Results are presented as mean number of responses for potato chips (trials 1–10) and M&M's (trials 11–13) for individuals assigned to (A) small (75 kcal; black circles) and large (225 kcal; white circles) portions of food and (B) who consumed the food throughout the duration of the experimental (black circles) and those who had to delay consumption until the end (white circles). Error bars represent the standard error of the mean. Results show that the rate of operant responding increases when the reinforcer is changed after the tenth 2-min interval (stimulus specificity). Results presented in (A) also show that the size the reinforcer does not influence the within-session pattern of operant responding, ruling out the idea that a variety of post-ingestive satiety factors (e.g., caloric density, stomach distension) contribute to the within-session changes in operant responding. Results presented in (B) also show that rate of operant responding decreases within sessions even when the reinforcer is not consumed, questioning the idea that post-ingestive satiety factors are the sole explanation for within-session changes in responding. From Temple et al. (2008, p. 394). Copyright 2007 by Elsevier Ltd. Reprinted with permission.

The predictions of this hypothesis were correct when Thompson and Spencer's (1966) empirical properties were used to define habituation. The following empirical properties of behavior underlying habituation were confirmed for within-session changes in operant responding. The within-session decreases in response rate take a negative exponential form (e.g., McSweeney, Hinson, & Cannon, 1996). Spontaneous recovery occurs between experimental sessions (e.g., Aoyama & McSweeney, 2001a; McSweeney & Johnson, 1994; Murphy, McSweeney, Kowal, McDonald, & Wiediger, 2006). The decreases in response rates are faster when stimuli are presented at faster than at slower rates (e.g., McSweeney, 1992). The decreases may be steeper for less, than for more intense, stimuli (Melville, Rue, Rybicki, & Weatherly, 1997). Long-term habituation occurs (e.g., McSweeney, 1992; McSweeney, Weatherly, & Swindell, 1995b). Dishabitators also lose their ability to restore habituated responding with their repeated presentation (Murphy et al., 2006).

Within-session changes in operant responding show several other properties that are also compatible with sensitization and habituation. For example, the within-session changes are produced by factors that accumulate within the experimental session, not by anticipation of factors that follow the session (McSweeney et al., 1995a). As with habituation (e.g., Thorpe, 1966), within-session response patterns are also highly general, occurring for a wide variety of species, responses, procedures, and reinforcers (e.g., McSweeney & Roll, 1993). McSweeney et al. (1996) reviewed some of the evidence for each of these characteristics.

In what follows, we discuss two additional properties of habituation (dishabituation, stimulus specificity) in detail. We have found them to be particularly useful in separating sensitization and habituation from the many alternative explanations for within-session changes in operant responding (e.g., fatigue, satiation, pharmacodynamic factors). Before beginning, we should note that our research has not yet specified the exact nature of the stimuli to which sensitization and habituation occur during operant conditioning procedures. We assume that animals sensitize and habituate to more than one stimulus and that these stimuli may differ for different animals. For example, rats and humans may sensitize and

habituate to the taste and smell of food because these stimuli play an important role in regulating their food consumption. Birds, which rely heavily on visual stimuli, may sensitize and habituate to the visual properties of food (e.g., its color). In addition to separating habituation from alternative explanations for within-session decreases in operant responding, tests for stimulus specificity may be used to identify the stimuli to which habituation occurs for each response. As will be described, if habituation occurs to one particular aspect of a reinforcer (e.g., color), then changing that aspect of the reinforcer should increase the operant responding supported by that reinforcer.

5. Dishabituation

Dishabituation refers to the fact that presenting a strong, different, or extra stimulus restores responsiveness to a habituated stimulus (e.g., Groves & Thompson, 1970). Consistent with dishabituation, Aoyama and McSweeney (2001b) found that response rates increased when the original stimulus was restored following a change in stimulus conditions. In their baseline condition, rats pressed a lever for food reinforcers (one Noyes pellet) delivered according to a fixed ratio (FR) 4 schedule in which every fourth lever press produced food. During experimental conditions, the stimulus situation changed for 3 min in the middle of the session. The following changes were introduced in different conditions: the lever was removed; the schedule was changed to an FR 6 schedule and two Noyes pellets were delivered whenever a reinforcer was earned; the schedule was changed to an FR 8 schedule; and an FR 2 schedule alternated with an FR 6 schedule. In all cases, response rate increased relative to baseline after the changed condition was removed and subjects once again responded on the FR 4 schedule. The increase in responding was observed regardless of whether rate of responding decreased (e.g., no lever) or increased (e.g., FR 8) relative to baseline while the stimulus was changed. It was also observed regardless of whether subjects obtained more (e.g., FR 6–2 pellets) or less (e.g., no lever) food relative to baseline while the stimulus was changed (see Fig. 3).

Aoyama and McSweeney's (2001b) results are consistent with habituation as an explanation for within-session decreases in responding because they show dishabituation. The results are not consistent with satiation or fatigue as explanations. If satiation produced the decreases in operant responding, then those decreases should have been steeper than baseline after experimental manipulations in which subjects earned more food than during baseline, and less steep than baseline after experimental manipulations in which subjects earned less food than during baseline. If fatigue produced the decreases in operant responding, then the decreases should have been steeper than baseline after manipulations in which subjects responded faster than during baseline. The decreases should have been less steep than baseline after manipulations in which subjects responded slower than during baseline. Instead, response rate increased after each stimulus change, consistent with habituation. (See McSweeney and Roll (1998), for similar results when pigeons pecked a key for food reinforcers.)

Murphy et al. (2006) reported dishabituation when rats responded for ethanol reinforcers. In their experiment, alcohol-prefering (P) rats responded on a variable interval (VI) 15-s schedule for reinforcers consisting of 5 s of access to a 10% ethanol

solution. A VI 15-s schedule presents approximately one reinforcer every 15 s on the average. Response rate for ethanol increased after a 60 dB tone was presented for 5 s (on for 1 s, off for 1 s, etc.) at 24 min and 55 s into the 50-min experimental session. A flashing light, presented similarly to the tone, did not alter response rate. However, Wistar rats, the parent strain of P rats, have poor visual acuity even relative to other strains of rats (e.g., Long-Evans; Prusky, Harker, Douglas, & Whishaw, 2002; see Fig. 4).

The increase in response rates that followed the tone is incompatible with pharmacodynamic factors, satiation, or motor impairment as explanations for the within-session decreases in responding for ethanol reinforcers. Presenting a tone should not alter any of these factors. The results are consistent with habituation. Presenting a tone should alter habituation through dishabituation. These results may also show the habituation of dishabituation, another fundamental property of behavior undergoing habituation (e.g., Thompson & Spencer, 1966). The increase in responding that followed the tone was larger in sessions 1–5 (top graph) than in sessions 6–10 (bottom graph). The decrease in the effectiveness of the tone may indicate that subjects were habituating to this dishabituator (see Fig. 4).

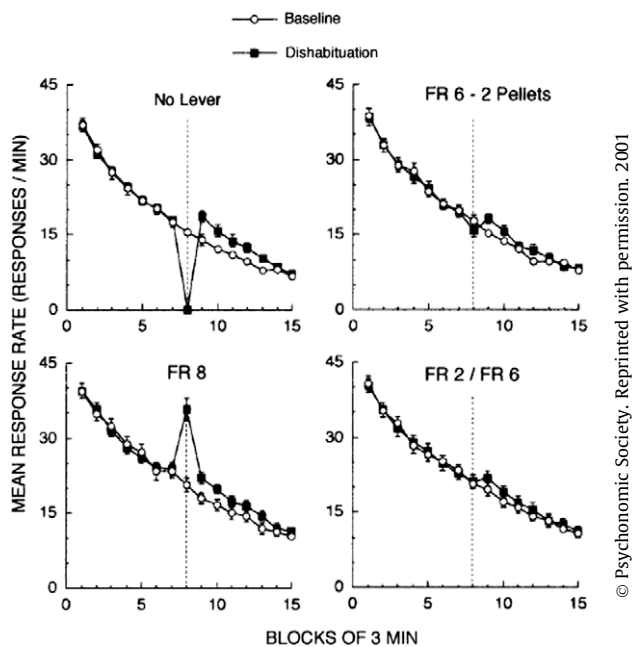


Fig. 3. Dishabituation of operant lever pressing in rats. Results are presented as rates of responding (responses/minute) during successive 3-min intervals in the session for the mean of all subjects responding during baseline (open circles) and experimental (filled squares) conditions. The dotted vertical line indicates the time at which the stimulus was briefly changed in the experimental conditions. In one experimental condition, the lever was withdrawn (top left graph). In another condition, lever pressing was reinforced on an FR 6 schedule and two Noyes pellets were delivered whenever a reinforcer was obtained (top right graph). In a third condition, the schedule was changed to an FR 8 (bottom left graph). In a final condition, two FR schedules (2 and 6) alternated (bottom right graph). Error bars represent the standard error of the mean. Results indicate that operant responding, supported by a constant reinforcer, increased after a change in the stimulus situation. Increases in responding occurred regardless of whether the rate of responding decreased (e.g., no lever, top left graph) or increased (e.g., FR 8, bottom left graph) relative to baseline during the stimulus change, challenging the idea that fatigue produces the within-session decreases in operant responding. Increases in responding also occurred regardless of whether subjects obtained more (e.g., FR 6–2 pellets, top right graph) or less (e.g., no lever, top left graph) food relative to baseline during the stimulus change, challenging the idea that non-habituation satiety variables produce the within-session decreases in operant responding. From Aoyama and McSweeney (2001b, p. 82).

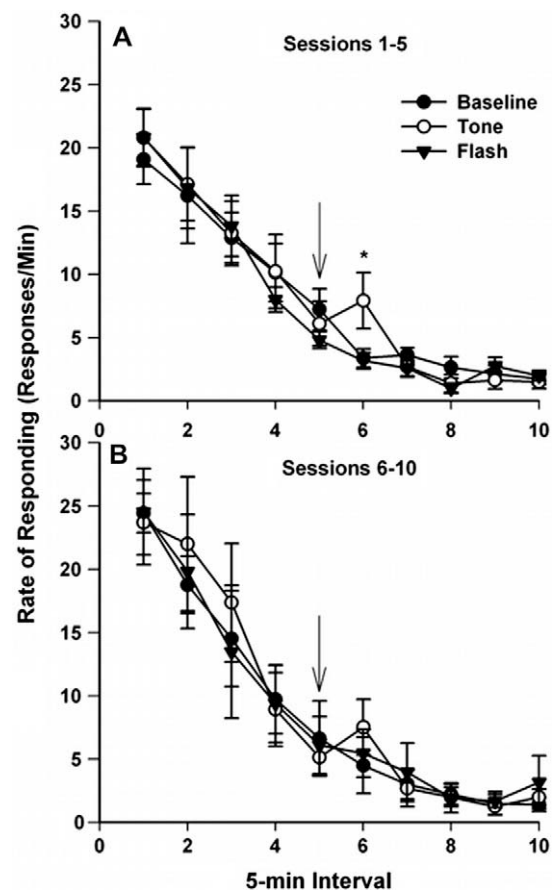


Fig. 4. Dishabituation of ethanol-reinforced lever pressing in alcohol-prefering rats. Results are presented as rates of responding (responses/minute) during successive 5-min intervals in the experimental session for the baseline (filled circles), tone (open circles), and flashing light (filled inverted triangles) conditions. The top graph presents the results from sessions 1 to 5. The bottom graph presents the results from sessions 6 to 10. Each function presents the mean of all subjects. Arrows indicate the time at which the tone or flashing light was presented. Error bars represent the standard error of the mean. Results show that operant responding for a constant reinforcer increased after the brief introduction of a tone (dishabituation), but not a light. From Murphy et al. (2006, p. 478).

Aoyama and McSweeney (2001a) observed dishabituation in wheel running. They studied a baseline condition in which rats ran in a running wheel for 30 min. During the experimental condition, a 5-s flashing light (house light off for 1 s, on for 1 s, etc.) was presented at 20 min and 55 s into the session. The rate of running increased for all subjects after the termination of the light (see Fig. 5).

The increase in running is incompatible with the idea that muscular fatigue produced the within-session decreases in wheel running. Muscular fatigue should not be altered by a flashing light. The increase in running is compatible with the idea that habituation produced the within-session decrease in running. Presenting the light should reduce habituation by producing dishabituation.

6. Stimulus specificity

Habituation is relatively specific to the presented stimulus. Therefore, habituation is disrupted by unpredicted changes in that stimulus (stimulus specificity; e.g., Swithers & Hall, 1994; Whitlow, 1975). The test for stimulus specificity resembles the test for dishabituation. Both tests involve an increase in responsiveness to a stimulus as a result of a change in the prior stimulus conditions. The two tests also differ, however. In a test for dishabituation, the change in stimulus conditions alters responsiveness to the original stimulus. In a test for stimulus specificity, responsiveness is greater to the new stimulus than to the immediately preceding original stimulus. The original stimulus is not reintroduced.

Epstein et al. (2003) allowed children to play a computer game for pieces of hamburger that were delivered on a VI 2-min schedule (a reinforcer was delivered approximately every 2 min on the average). After 10 or 11 2-min trials (in different conditions), the reinforcer was changed to apple pie. The stimulus changed on two different trials to ensure that the changes in responding were not produced by variables related to the passage of time.

Consistent with a violation of stimulus specificity, changing the food increased the children's rate of operant responding. These increases in response rates are incompatible with satiety variables

(other than habituation) as explanations for the within-session decreases in operant responding. Stomach distension, blood glucose levels, CCK, etc. should only increase with additional food, producing further decreases in response rates (see Fig. 6. See also Temple et al. (2008) (Fig. 2), for similar results for children responding for potato chips followed by chocolate candy.).

Perhaps because of stimulus specificity, responding that is governed by habituation also shows variety effects. That is, habituation occurs more slowly to stimuli that are presented in a variable, than in a fixed, manner (e.g., Broster & Rankin, 1994). Aoyama and McSweeney (2001b) showed that within-session decreases in operant responding were steeper when rats pressed levers for food reinforcers (one Noyes pellet) delivered by four different FR schedules than by matched variable ratio (VR) schedules. FR schedules present reinforcers after a fixed number of responses; VR schedules present reinforcers after a variable number of responses (see Fig. 7).

Again, these results are incompatible with satiety variables other than habituation as explanations for within-session decreases in operant responding. The matched FR and VR schedules provided similar amounts of food. Therefore, stomach distension, blood glucose levels, CCK, etc. should have been comparable for the two schedules. In contrast, habituation predicts the difference in the within-session response patterns. Habituation should be slower when stimuli are presented in a variable, rather than a fixed, pattern.

Ernst and Epstein (2002) allowed human participants to play a computer game for food delivered on a VR 32 schedule (a reinforcer was presented after 32 responses on the average). In one condition, the food was constant across trials (pieces of turkey sandwich). In the other condition, the food varied among three different foods that provided the same number of calories as the constant food (pieces of turkey or roast beef sandwiches, potato chips). Within-session decreases in responding were steeper for those given the same food each time a reinforcer was delivered than for those given different, but isocaloric, foods. If the satiety variable, caloric density, produced the within-session decreases in operant responding, those decreases should have been

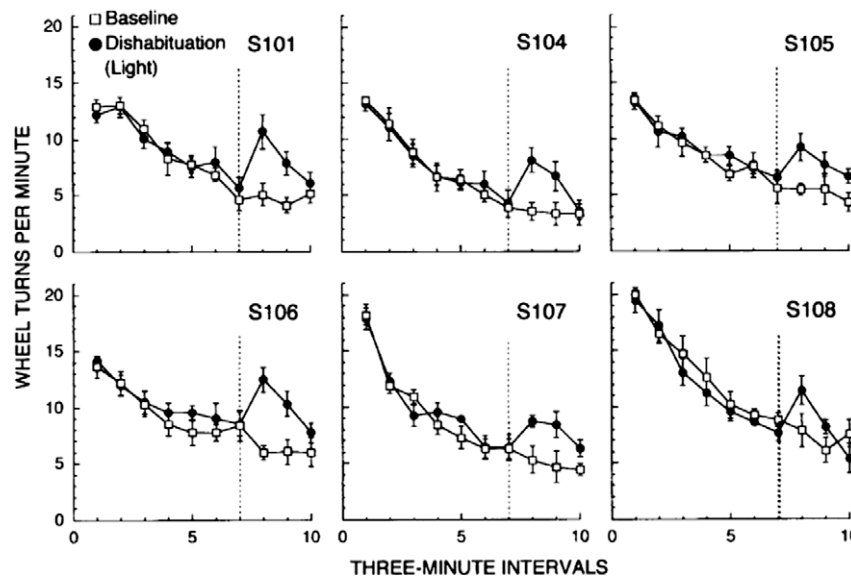


Fig. 5. Dishabituation of wheel running in rats. Results are presented as the number of wheel turns per minute during successive 3-min intervals in the session in the baseline (open squares) and experimental (filled circles) conditions. Each graph presents the results for an individual subject. The vertical dashed lines indicate the time at which the flashing light was presented. Error bars indicate the standard error of the mean. Results show that the rate of wheel running increased after the brief introduction of a light (dishabituation). From Aoyama and McSweeney (2001a, p. 294).

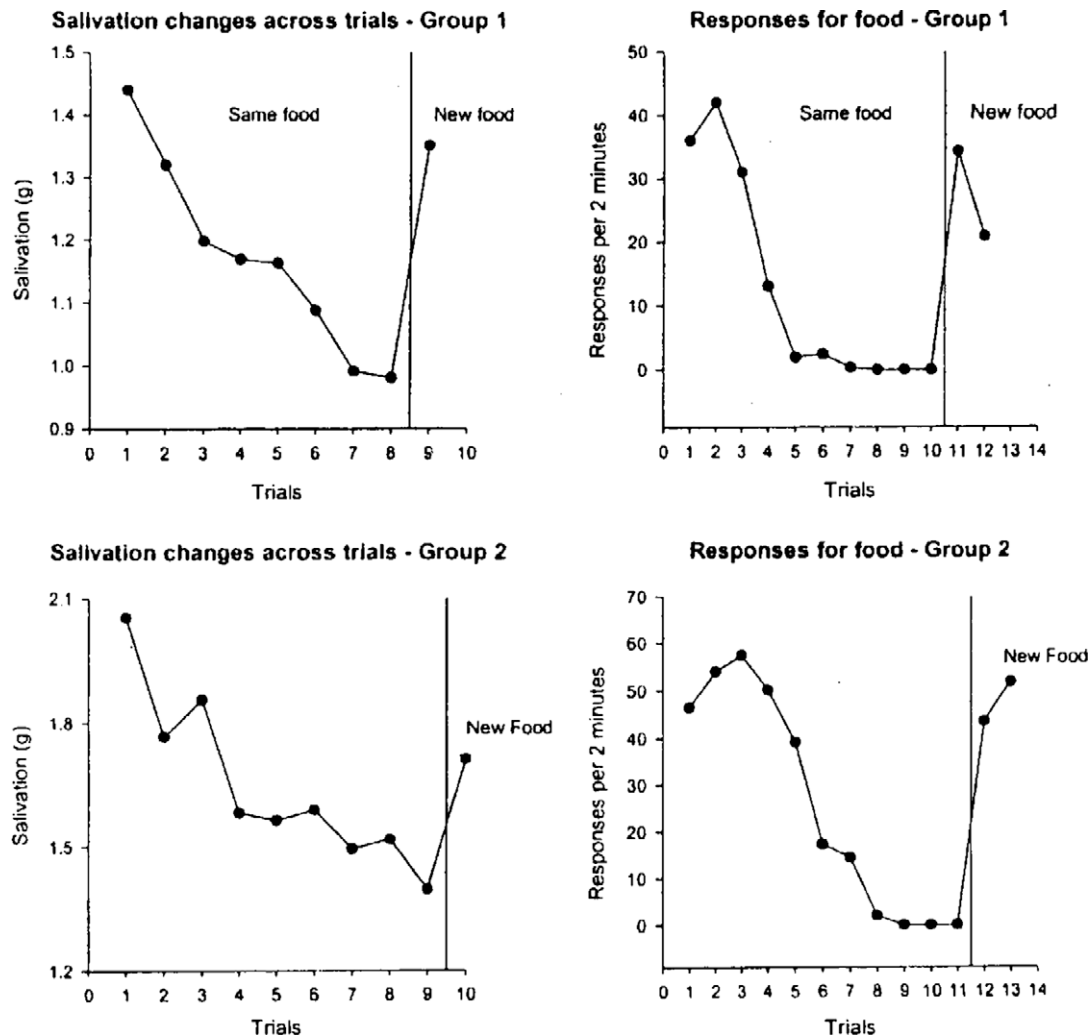


Fig. 6. Stimulus specificity of operant responding in children. Results are presented as the amount of salivation (left graphs) and the rate of operant responding (right graphs) for the mean of all children exposed to, or responding for, hamburger and then apple pie during successive 2-min trials. The trial on which the reinforcer changed from hamburger to apple pie is indicated by a vertical line. The results show that salivation (left graphs) and operant responding for food (right graphs) increased after the reinforcer changed (stimulus specificity). The similarity in the within-session changes in salivation and operant responding may suggest that a common variable (e.g., habituation) produces both changes. Nevertheless, this similarity should not be overly interpreted because it illustrates only a correlation between the strengths of the two responses. From Epstein et al. (2003, p. 286). Copyright 2003 by Elsevier Ltd. Reprinted with permission.

similar for isocaloric sandwiches. Instead, variety slowed the within-session decrease in responding as predicted by habituation (see Fig. 8. See Melville et al. (1997), for similar results for rats pressing levers for food.).

Murphy, McSweeney, and Kowal (2007) allowed P rats to press a lever for ethanol reinforcers (5-s access to a 10% ethanol solution). In different conditions, rats responded on fixed interval (FI) or matched VI schedules (7.5-s, 15-s, and 30-s). FI schedules allow access to reinforcers after a fixed period of time; VI schedules allow access to reinforcers after a variable period of time. Response rate decreased more slowly within the session when subjects responded on the VI schedules than on the matched FI schedules. These results are incompatible with explanations for the within-session decreases in responding in terms of pharmacodynamic factors, satiety, or motor impairment produced by ethanol. The matched FI and VI schedules provided similar amounts of ethanol. Therefore, each of these variables should have been comparable for the two schedules. In contrast, habituation predicts the difference in the within-session response patterns. Habituation should be slower when stimuli are presented in a variable than a fixed manner (see Fig. 9).

7. Implications

Arguing that habituation may occur to biologically important as well as to neutral stimuli, and for voluntary as well as for reflexive behavior, has many implications. For example, sensitization and habituation to the reinforcer might contribute to the understanding of such poorly understood phenomena in operant conditioning as multiple-schedule behavioral contrast (McSweeney & Weatherly, 1998), extinction (e.g., McSweeney & Swindell, 2002), and results that are often attributed to behavioral economics (McSweeney & Swindell, 1999a; McSweeney, Swindell, & Weatherly, 1996b).

Arguing that sensitization and habituation alter the effectiveness of reinforcers has implications for the use of reinforcers in practice. For example, an understanding of the characteristics of habituation can be used to strengthen a reinforcer that is too weak (e.g., food used to change the behavior of an autistic child) or to weaken a reinforcer that is too strong (e.g., food for someone who is obese. See Murphy, McSweeney, Smith, and McComas (2003), for a discussion of applied implications.).

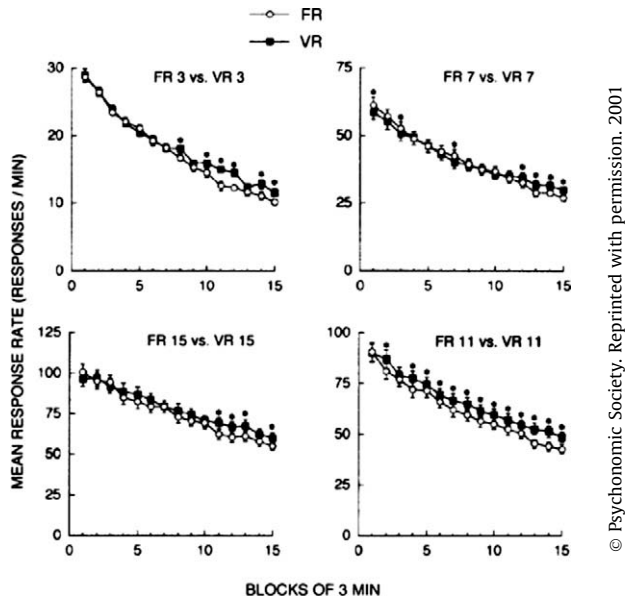


Fig. 7. Variety effects in rats. Results are presented as rates of responding (responses/minute) during successive 3-min intervals in a 45-min session for the mean of all subjects. In different conditions, rats pressed levers for food reinforcers delivered on FR 3 vs. VR 3 (top left graph), FR 7 vs. VR 7 (top right graph), FR 11 vs. VR 11 (bottom right graph), and FR 15 vs. VR 15 (bottom left graph) schedules. Open circles present the results for the FR schedules; filled squares, the results for the VR schedules. Error bars represent the standard error of the mean. Asterisks indicate when subjects responded at significantly different rates during the matched FR and VR schedules. The results show that within-session decreases in operant responding are slower when food reinforcers are presented in a variable (filled squares) rather than in a fixed (open circles) manner. From Aoyama and McSweeney (2001b, p. 86).

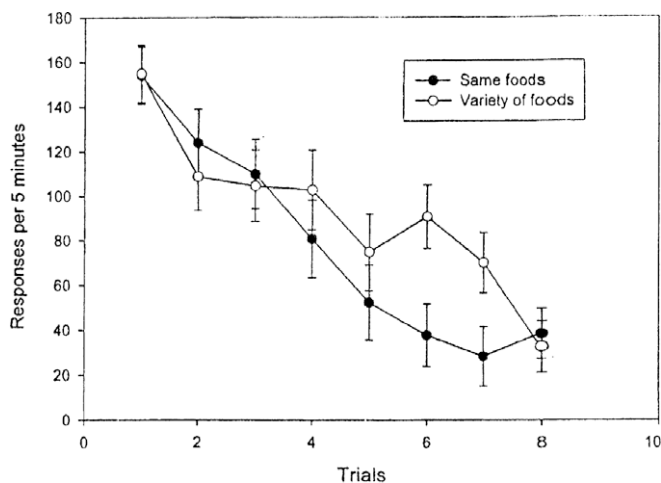


Fig. 8. Variety effects in humans. Results are presented as the number of responses per 5-min trial as a function of successive trials for the mean of all human subjects responding for turkey sandwiches (filled circles) or for three different isocaloric foods (open circles). Error bars represent the standard error of the mean. Results show that human operant responding for food reinforcers decreases more slowly within the experimental session when a variety of reinforcers (open circles) is presented than when a fixed reinforcer (filled circles) is presented. From Ernst and Epstein (2002, p. 229). Copyright 2002 by Elsevier Ltd. Reprinted with permission.

Arguing that habituation occurs to biologically important stimuli implies that habituation may contribute to the weakening of the unconditioned stimulus (US) that is often observed in classical conditioning after repeated US presentations. The size of the unconditioned responses (UR) may decrease during conditioning (conditioned diminution of the UR; see, e.g., Kimmel, 1966). Classical conditioning may be weaker when several USs are presented

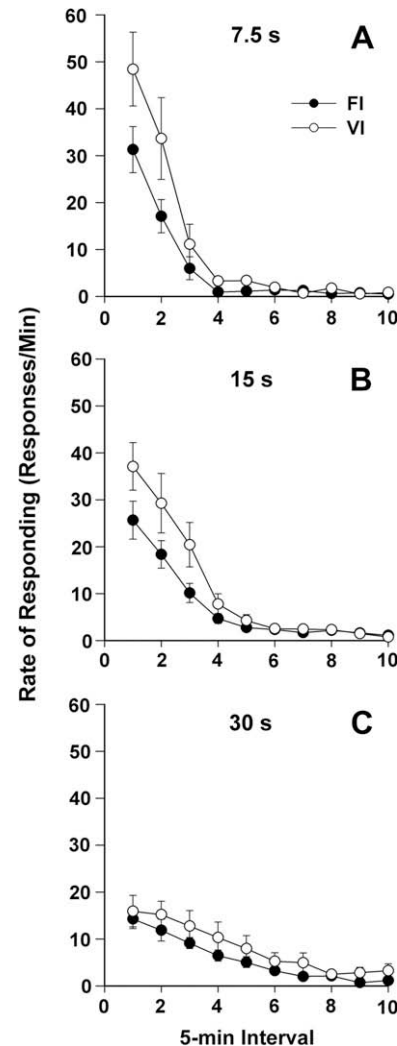


Fig. 9. Variety effects in alcohol-preferring rats. Results are presented as rates of responding (responses/min) during the fixed- (filled circles) and variable- (open circles) interval schedules of ethanol reinforcement during successive 5-min intervals for the 7.5-s (A), 15-s (B), and 30-s (C) inter-reinforcer intervals. Each function represents the mean of all subjects responding during the last five sessions of each condition. Error bars represent the standard error of the mean. Results show that operant responding decreases more slowly within sessions when reinforcers are presented after variable (open circles), rather than after fixed (closed circles), periods of time. From Murphy et al. (2007, p. 118).

before conditioning begins than when they are not (the US pre-exposure effect; see, e.g., Mis & Moore, 1973). Conditioning is usually weaker when a US is presented immediately before a conditioned stimulus (CS)-US pairing than when it is not (priming by the US; see, e.g., Terry, 1976). Conditioning is usually also weaker when a CS is paired with a weak US before it is paired with a strong US than when it is not (the Hall-Pearce Effect; Hall & Pearce, 1979). Although it has been argued that habituation may contribute to these effects (see, e.g., Randich & LoLordo, 1979; Wagner, 1976), this argument is contradicted by the habituation literature if habituation is restricted to neutral stimuli. The US in classical conditioning is usually a biologically significant stimulus (e.g., food).

Similar changes also occur in the effectiveness of the CS in classical conditioning with its repeated presentation. The salience of the CS may change during conditioning (see, e.g., Pearce & Hall, 1980). Classical conditioning is usually weaker when several CSs are presented before conditioning begins than when they are not (latent inhibition; see, e.g., Lubow, 1989). Conditioning is usually weaker when a CS is presented immediately before a CS-US pairing

than when it is not (priming by the CS; see, e.g., Pfautz & Wagner, 1976). In addition, habituation to both the CS and the US may contribute to learned irrelevance (see, e.g., Mackintosh, 1973), the finding that conditioning is retarded following uncorrelated presentations of the CS and US (see, e.g., Bonardi & Hall, 1996). The similarities in the changes in the effectiveness of the CS and US with their repeated presentation suggest that these changes are produced by a common variable or variables. Habituation provides one such potential variable only if its domain is expanded to include biologically significant stimuli.

Finally, habituation may also contribute to the regulation of “motivated” or goal-directed behaviors (e.g., feeding, drinking, sex, aggression, drug taking). Early theories attempted to explain all of these behaviors through a common process (e.g., hedonism, instincts, drives, incentives). Each of these common-process theories failed, probably partly because motivated behaviors appear to be a heterogeneous class. The class includes some basic biological behaviors (feeding, drinking, sexual and maternal behavior, aggression, escape), as well as behaviors that are directed at more arbitrary stimuli (e.g., exploration, curiosity, power and money seeking). Theories that dealt well with biologically important behaviors (e.g., instincts) tended to fail for more arbitrary behaviors and vice versa.

When these general-process explanations for motivated behaviors failed, the control of different behaviors was attributed to different theoretical variables. For example, termination of ingestion (e.g., eating, drinking) is usually attributed to satiation (e.g., Bizo, Bogdanov, & Killeen, 1998); termination of energetic responding (e.g., running), to fatigue (e.g., Belke, 1997); termination of cognitive behaviors (e.g., studying), to the waning of attention (e.g., Hinson & Tennison, 1999); termination of drug taking, to pharmacodynamic factors (e.g., Ahmed & Koob, 1999). Unfortunately, the use of different terms to describe different behaviors is unparsimonious and is contradicted by the many common characteristics possessed by different motivated behaviors (McSweeney & Swindell, 1999b).

If the domain of habituation is expanded to include biologically important stimuli, then habituation might provide a general-process contributor to the regulation of many motivated behaviors. According to this explanation, each motivated behavior is maintained by a reinforcer (e.g., food for eating; the drug for drug taking; money for money seeking). The behavior stops when habituation weakens the reinforcer so that the reinforcer no longer supports behavior directed towards it. The behavior resumes when spontaneous recovery sufficiently restores the effectiveness of the reinforcer so that it again supports behavior. It should be noted that this model does not argue that habituation is the sole regulator of all motivated behavior. Instead, the model argues that habituation is one of several variables that contribute to the regulation of many motivated behaviors. (See McSweeney and Swindell (1999b), for a more detailed discussion of this idea.)

The habituation model is similar to the most popular and persistent general-process explanation for motivated behavior: homeostasis. Models based on habituation and homeostasis both postulate that motivated behaviors are governed by a negative feedback loop in which engaging in the behavior leads to the termination of that behavior. Homeostatic models argue that the behavior is triggered by a physiological deficit that accumulates over time (e.g., feeding is triggered by a decrease in blood sugar levels). Performing the behavior corrects the deficit (e.g., eating increases blood sugar) and terminates the behavior. The habituation model argues that a motivated behavior occurs when spontaneous recovery sufficiently restores the effectiveness of the reinforcer that supports that behavior. Performing the behavior terminates it through habituation to its supporting reinforcer.

An explanation in terms of habituation avoids two major problems encountered by homeostasis. First, homeostatic models postulate a specific physiological basis for the negative feedback loop that governs each motivated behavior (e.g., fluctuations in blood sugar for eating; fluctuations in intra- or extracellular fluid for drinking). Many motivated behaviors, such as exploration, achievement, or money seeking are not obviously triggered by the accumulation of a physiological deficit or terminated by the restoration of that deficit. Second, even behaviors that appear to be homeostatic (e.g., feeding, drinking) are not when they are examined carefully. For example, animals may drink long before there is a physiological need to do so and stop drinking before systemic changes repair the deficit (Blass & Hall, 1976; Fitzsimons & Le Magnen, 1969; Phillips, Rolls, Ledingham, & Morton, 1984). The habituation model avoids the first problem because it does not postulate a different physiological mechanism to govern each motivated behavior. Instead, it requires only that the goal object function as a reinforcer to which habituation can occur, an intuitively appealing and parsimonious assumption. The habituation model avoids the second problem because habituation can occur quickly. It does not require time for the correction of an underlying physiological deficit.

The habituation model is parsimonious because it appeals to only two processes (habituation, reinforcement). The model is highly general because habituation occurs for many, if not all, stimuli and species of animal. The model is also testable because habituation and reinforcement are relatively well understood. Therefore, the literatures devoted to them can be used to make relatively specific predictions about motivated behaviors. McSweeney and Swindell (1999b) followed this approach and reviewed the literature on motivated behavior. They showed that motivated behaviors shared many of the properties of behaviors undergoing habituation. (See also Raynor and Epstein (2001).) Some of these predictions were surprising because they were counterintuitive. For example, as predicted by habituation, motivated behaviors such as feeding are strengthened by variety (violations of stimulus specificity) and by the presence of irrelevant stimuli (dishabituation, sensitization). Once an animal stops eating a food, giving a nibble of that food restores eating (sensitization).

The habituation model also has many potential practical implications that differ from those offered by traditional models of motivated behavior. To take eating as an example, the traditional model of the regulation of eating emphasizes the role of post-ingestive factors (e.g., energy intake). To reduce food consumption, a person should eat fewer and lower calorie foods. Although the habituation model does not deny the importance of post-ingestive factors, it emphasizes the importance of other variables such as the sensory properties of the food and the sensory environment in which the person eats. To avoid over-consumption, a person should eat relatively constant foods. As noted, variety in food reduces the rate of habituation and therefore, can lead to over-consumption (e.g., Ernst & Epstein, 2002; Raynor & Epstein, 2001). To reduce intake, people should not eat in places with many environmental distracters (e.g., in front of the television, in a noisy bar; Temple, Giacomelli, Kent, Roemmich, & Epstein, 2007). These unpredictable stimuli will slow habituation by acting as dishabitators. A person should put away snacks once they have stopped eating. A nibble of an easily available food may produce sensitization, which will briefly increase the reinforcing value of the food and lead to the resumption of eating. Because consumption is not necessary to produce habituation to food (see e.g., Fig. 2), exposure to the sight and/or odor of food before consumption might decrease later consumption of that food by producing habituation to some of the food's sensory properties. It should be noted that this exposure to the sight and/or taste of food should be long enough to ensure that habituation, rather than sensitization, occurs. The

habituation model also suggests that it would be worthwhile to compare overweight and normal weight subjects on such variables as the strength of food as a reinforcer (e.g., Saelens & Epstein, 1996), rate of habituation (e.g., Epstein, Paluch, & Coleman, 1996), amount of sensitization, rate of spontaneous recovery, sensory properties of typically eaten foods, sensory environments in which eating typically occurs, etc. Differences in any or all of these variables might support the differences in consumption that contribute to differences in weight.

The results cited in the present paper provide evidence that habituation contributes to the cessation of three motivated behaviors: feeding (usually attributed to satiation), running (usually attributed to fatigue), and drug taking (usually attributed to pharmacodynamic factors). It remains to be seen whether habituation contributes to the regulation of additional motivated behaviors. However, the unusual and relatively specific predictions of habituation suggest that this hypothesis is worth testing.

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References

- Ahmed, S. H., & Koob, G. F. (1999). Long-lasting increase in the set point for cocaine self-administration after escalation in rats. *Psychopharmacology*, 146, 303–312.
- Aoyama, K., & McSweeney, F. K. (2001a). Habituation contributes to within-session changes in free wheel running. *Journal of the Experimental Analysis of Behavior*, 76, 289–302.
- Aoyama, K., & McSweeney, F. K. (2001b). Habituation may contribute to within-session decreases in responding under high-rate schedules of reinforcement. *Animal Learning & Behavior*, 29, 79–91.
- Belke, T. W. (1997). Running and responding reinforced by the opportunity to run: Effect of reinforcer duration. *Journal of the Experimental Analysis of Behavior*, 67, 337–351.
- Bizo, L. A., Bogdanov, S. V., & Killeen, P. R. (1998). Satiation causes within-session decreases in instrumental responding. *Journal of Experimental Psychology: Animal Behavior Processes*, 24, 439–452.
- Blass, E. M., & Hall, W. G. (1976). Drinking termination: Interactions among hydration, orogastric, and behavioral controls in rats. *Psychological Review*, 83, 356–374.
- Bonardi, C., & Hall, G. (1996). Learned irrelevance: No more than the sum of CS and US preexposure effects? *Journal of Experimental Psychology: Animal Behavior Processes*, 22, 183–191.
- Broster, B. S., & Rankin, C. H. (1994). Effects of changing interstimulus interval during habituation in *Caenorhabditis elegans*. *Behavioral Neuroscience*, 108, 1019–1029.
- Epstein, L. H., Paluch, R., & Coleman, K. J. (1996). Differences in salivation to repeated food cues in obese and nonobese women. *Psychosomatic Medicine*, 58, 160–164.
- Epstein, L. H., Rodefer, J. S., Wisniewski, L., & Caggia, A. R. (1992). Habituation and dishabituation of human salivary response. *Physiology & Behavior*, 51, 945–950.
- Epstein, L. H., Saad, F. G., Handley, E. A., Roemmich, J. N., Hawk, L. W., & McSweeney, F. K. (2003). Habituation of salivation and motivated responding for food in children. *Appetite*, 41, 283–289.
- Ernst, M. M., & Epstein, L. H. (2002). Habituation of responding for food in humans. *Appetite*, 38, 224–234.
- Fitzsimons, T. J., & Le Magnen, J. (1969). Eating as a regulatory control of drinking in the rat. *Journal of Comparative and Physiological Psychology*, 67, 273–283.
- Groves, P. M., & Thompson, R. F. (1970). Habituation: A dual-process theory. *Psychological Review*, 77, 419–450.
- Hall, G., & Pearce, J. M. (1979). Latent inhibition of a CS during CS-US pairings. *Journal of Experimental Psychology: Animal Behavior Processes*, 5, 31–42.
- Harris, J. D. (1943). Habitatory response decrement in the intact organism. *Psychological Bulletin*, 40, 385–422.
- Hinson, J. M., & Tennyson, L. R. (1999). Within-session analysis of visual discrimination. *Journal of the Experimental Analysis of Behavior*, 72, 385–405.
- Kimmel, H. D. (1966). Inhibition of the unconditioned response in classical conditioning. *Psychological Review*, 73, 232–240.
- Lubow, R. E. (1989). *Latent inhibition and conditioned attention theory*. New York: Cambridge University Press.
- Mackintosh, N. J. (1973). Stimulus selection: Learning to ignore stimuli that predict no change in reinforcement. In R. A. Hinde & J. S. Hinde (Eds.), *Constraints on learning: Limitations and predispositions* (pp. 75–96). Oxford: Academic Press.
- McSweeney, F. K. (1992). Rate of reinforcement and session duration as determinants of within-session patterns of responding. *Animal Learning & Behavior*, 20, 160–169.
- McSweeney, F. K., Hinson, J. M., & Cannon, C. B. (1996). Sensitization–habituation may occur during operant conditioning. *Psychological Bulletin*, 120, 256–271.
- McSweeney, F. K., & Johnson, K. S. (1994). The effect of time between sessions on within-session patterns of responding. *Behavioural Processes*, 31, 207–217.
- McSweeney, F. K., & Roll, J. M. (1993). Responding changes systematically within sessions during conditioning procedures. *Journal of the Experimental Analysis of Behavior*, 60, 621–640.
- McSweeney, F. K., & Roll, J. M. (1998). Do animals satiate or habituate to repeatedly presented reinforcers? *Psychonomic Bulletin & Review*, 5, 428–442.
- McSweeney, F. K., Roll, J. M., & Weatherly, J. N. (1994). Within-session changes in responding during several simple schedules. *Journal of the Experimental Analysis of Behavior*, 62, 109–132.
- McSweeney, F. K., & Swindell, S. (1999a). Behavioral economics and within-session changes in responding. *Journal of the Experimental Analysis of Behavior*, 72, 355–371.
- McSweeney, F. K., & Swindell, S. (1999b). General-process theories of motivation revisited: The role of habituation. *Psychological Bulletin*, 125, 437–457.
- McSweeney, F. K., & Swindell, S. (2002). Common processes may contribute to extinction and habituation. *Journal of General Psychology*, 129, 364–400.
- McSweeney, F. K., Swindell, S., & Weatherly, J. N. (1996a). Within-session changes in adjunctive and instrumental responding. *Learning and Motivation*, 27, 408–427.
- McSweeney, F. K., Swindell, S., & Weatherly, J. N. (1996b). Within-session changes in responding during concurrent schedules with different reinforcers in the components. *Journal of the Experimental Analysis of Behavior*, 66, 369–390.
- McSweeney, F. K., & Weatherly, J. N. (1998). Habituation to the reinforcer may contribute to multiple-schedule behavioral contrast. *Journal of the Experimental Analysis of Behavior*, 69, 199–221.
- McSweeney, F. K., Weatherly, J. N., & Roll, J. M. (1995). Within-session changes in responding during concurrent schedules that employ two different operanda. *Animal Learning & Behavior*, 23, 237–244.
- McSweeney, F. K., Weatherly, J. N., Roll, J. M., & Swindell, S. (1995). Within-session patterns of responding when the operandum changes during the session. *Learning and Motivation*, 26, 403–420.
- McSweeney, F. K., Weatherly, J. N., & Swindell, S. (1995a). Prospective factors contribute little to within-session changes in responding. *Psychonomic Bulletin & Review*, 2, 234–238.
- McSweeney, F. K., Weatherly, J. N., & Swindell, S. (1995b). Within-session changes in key and lever pressing for water during several multiple variable-interval schedules. *Journal of the Experimental Analysis of Behavior*, 64, 75–94.
- McSweeney, F. K., Weatherly, J. N., & Swindell, S. (1996a). Reinforcer value may change within experimental sessions. *Psychonomic Bulletin & Review*, 3, 372–375.
- McSweeney, F. K., Weatherly, J. N., & Swindell, S. (1996b). Within-session changes in responding during delayed matching-to-sample and discrimination procedures. *Animal Learning & Behavior*, 24, 290–299.
- Melville, C. L., Rue, H. C., Rybicki, L. R., & Weatherly, J. N. (1997). Altering reinforcer variety or intensity changes the within-session decrease in responding. *Learning and Motivation*, 28, 609–621.
- Melville, C. L., Rybicki, L. R., & Kamrani, B. (1996). Within-session responding as a function of force required for lever press. *Behavioural Processes*, 37, 217–224.
- Melville, C. L., & Weatherly, J. N. (1996). Within-session patterns of responding when rats run in a T-maze. *Behavioural Processes*, 38, 89–102.
- Millenson, J. R. (1967). *Principles of behavioral analysis*. New York: Macmillan.
- Mis, F. W., & Moore, J. W. (1973). Effect of preacquisition UCS exposure on classical conditioning of the rabbit's nictitating membrane response. *Learning and Motivation*, 4, 108–114.
- Mook, D. G. (1996). *Motivation: The organization of action* (2nd ed.). New York: W. W. Norton.
- Murphy, E. S., McSweeney, F. K., & Kowal, B. P. (2007). Motivation to consume alcohol in rats: The role of habituation. In P. W. O'Neal (Ed.), *Motivation of health behavior* (pp. 111–126). Hauppauge, NY: Nova Science.
- Murphy, E. S., McSweeney, F. K., Kowal, B. P., McDonald, J., & Wiediger, R. V. (2006). Spontaneous recovery and dishabituation of ethanol-reinforced responding in alcohol-preferring rats. *Experimental and Clinical Psychopharmacology*, 14, 471–482.
- Murphy, E. S., McSweeney, F. K., Smith, R. G., & McComas, J. J. (2003). Dynamic changes in reinforcer effectiveness: Theoretical, methodological, and practical implications for applied research. *Journal of Applied Behavior Analysis*, 36, 421–438.
- Pfautz, P. L., & Wagner, A. R. (1976). Transient variations in responding to Pavlovian conditioned stimuli have implications for the mechanisms of “priming”. *Animal Learning & Behavior*, 4, 107–112.
- Pearce, J. M., & Hall, G. (1980). A model for Pavlovian learning: Variations in the effectiveness of conditioned but not of unconditioned stimuli. *Psychological Review*, 87, 532–552.
- Phillips, P. A., Rolls, B. J., Ledingham, J. G. G., & Morton, J. J. (1984). Body fluid changes, thirst and drinking in man during free access to water. *Physiology & Behavior*, 33, 357–363.
- Prusky, G. T., Harker, K. T., Douglas, R. M., & Whishaw, I. Q. (2002). Variation in visual acuity within pigmented, and between pigmented and albino rat strains. *Behavioural Brain Research*, 136, 339–348.

- Randich, A., & LoLordo, V. M. (1979). Associative and nonassociative theories of the UCS preexposure phenomenon: Implications for Pavlovian conditioning. *Psychological Bulletin*, 86, 523–548.
- Raynor, H. A., & Epstein, L. H. (2001). Dietary variety, energy regulation, and obesity. *Psychological Bulletin*, 127, 325–341.
- Roll, J. M., & McSweeney, F. K. (1997). Within-session changes in operant responding when gerbils (*Meriones unguiculatus*) serve as subjects. *Current Psychology*, 15, 340–345.
- Roll, J. M., McSweeney, F. K., Johnson, K. S., & Weatherly, J. N. (1995). Satiety contributes little to within-session decreases in responding. *Learning and Motivation*, 26, 323–341.
- Saelens, B. E., & Epstein, L. H. (1996). Reinforcing value of food in obese and non-obese women. *Appetite*, 27, 41–50.
- Solomon, R. L., & Corbit, J. D. (1974). An opponent-process theory of motivation: I. Temporal dynamics of affect. *Psychological Review*, 81, 119–145.
- Swithers, S. E., & Hall, W. G. (1994). Does oral experience terminate ingestion? *Appetite*, 23, 113–138.
- Temple, J. L., Giacomelli, A. M., Kent, K. M., Roemmich, J. N., & Epstein, L. H. (2007). Television watching increases motivated responding for food and energy intake in children. *American Journal of Clinical Nutrition*, 85, 355–361.
- Temple, J. L., Giacomelli, A. M., Roemmich, J. N., & Epstein, L. H. (2008). Habituation and within session changes in motivated responding for food in children. *Appetite*, 50, 390–396.
- Terry, W. S. (1976). Effects of priming unconditioned stimulus representation in short-term memory on Pavlovian conditioning. *Journal of Experimental Psychology: Animal Behavior Processes*, 2, 354–369.
- Thompson, R. F., & Spencer, W. A. (1966). Habituation: A model phenomenon for the study of neuronal substrates of behavior. *Psychological Review*, 73, 16–43.
- Thorpe, W. H. (1966). *Learning and instinct in animals*. Cambridge, MA: Harvard University Press.
- Wagner, A. R. (1976). Priming in STM: An information-processing mechanism for self-generated or retrieval-generated depression in performance. In T. J. Tighe & R. N. Leaton (Eds.), *Habituation: Perspectives from child development, animal behavior and neurophysiology* (pp. 95–128). New York: Wiley.
- Weatherly, J. N., McSweeney, F. K., & Swindell, S. (1995). On the contributions of responding and reinforcement to within-session patterns of responding. *Learning and Motivation*, 26, 421–432.
- Whitlow, J. W. Jr., (1975). Short-term memory in habituation and dishabituation. *Journal of Experimental Psychology: Animal Behavior Processes*, 1, 189–206.